

Qian HU

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EDUCATION & RESEARCH

Australian National University

2025.12 - Present **Visiting PhD student** Supervised by Daniel Noble
Project: Thermal plasticity of avian embryonic development

Beijing Forestry University

2021 - Present **PhD student** Supervised by Jianqiang Li & Jiliang Xu
Project: Temperature-driven reproductive strategies, embryonic development, and adaptive evolution in uniparentally and biparentally incubating birds

2017 - 2021 **Master** Supervised by Jianqiang Li
Project: A comparative study of different incubation modes and their adaptability of two *Aegithalos* species

2013 - 2017 **Bachelor** Supervised by Gangming Zhang
Project: Phylogenetic analysis of the Asian “Cheilanthes” —— Based on two nuclear DNA sequences

RESEARCH FOCUS

- **Avian Reproduction & Parental Care**
Studying how birds regulate reproduction through reproductive strategies, parental investment, and incubation behaviour, and how environmental conditions shape parental care strategies.
- **Incubation Behaviour & Thermal Ecology**
Studying how birds regulate the thermal environment of developing embryos through incubation behaviour, and how temperature influences incubation investment, incubation strategies, and reproduction.
- **Embryonic Development & Developmental Plasticity**
Studying how temperature and other early-life environmental conditions shape embryonic development, phenotypes, and survival, and how developmental responses contribute to adaptation to environmental change.
- **Comparative Physiology & Evolutionary Ecology**
Using comparative and evolutionary approaches to investigate how physiological and behavioural responses to environmental variation are shaped by reproductive strategies, ecological conditions, and evolutionary history.

SKILLS

- **Field and lab skills:** Nest searching & monitoring, Breeding behaviour monitoring, Bird banding, Wildlife photography, PCR, Embryonic heart-rate monitoring (Buddy), Songbird blood sampling for hormone analysis.
- **Computer skills:** Windows, R (*lme4*, *MuMIn*, *RMark*, *MCMCglmm*, *ggplot2*, *metafor*, *orchaRd*, etc.), SPSS, Microsoft Office, Adobe Creative Suite (Photoshop, Illustrator, Premiere).
- **Language:** Mandarin, English.

PUBLICATIONS

1. **Hu, Q.**, Wang, Y., Wang, H., Wang, Y., Xia, X., Xu, J., & Li, J. (2026). Parental care and bird embryonic metabolism: A comparison between uniparentally incubating Silver-throated Tits and biparentally incubating Black-throated Tits. *Functional Ecology*, 40(4), 800-815.
2. **Hu, Q.**, Xu, J., & Li, J. (2026). Ecological and Life-History Correlates of Incubation Attentiveness Differ Between Female-Only and Biparentally Incubating Passerine Birds. *Global Ecology and Biogeography*, 35(3), e70225.
3. **Hu, Q.**, Wang, Y., Yu, G., Lv, L., Wang, P., Wen, Y., ... & Li, J. (2025). Biparentally incubating *Aegithalos concinnus* (Black-throated Tit) provide a more stable nest thermal environment than uniparentally incubating *A. glaucogularis* (Silver-throated Tit). *Ornithology*, 142(4), ukaf025.
4. **Hu, Q.**, Wang, Y., Yu, G., Lv, L., Wang, P., Wen, Y., ... & Li, J. (2024). The effect of ambient temperature on bird embryonic development: a comparison between uniparental incubating silver-throated tits and biparental incubating black-throated tits. *Journal of Avian Biology*, 2024(5-6), e03168.
5. 胡骞, 文野, 吕磊, 王鹏程, 李建强, & 徐基良. (2023). 研究活动是否影响鸟类的巢存活率?——以红头长尾山雀为例. *生物多样性*, 31(6), 22570.
6. **Hu, Q.**, Wen, Y., Yu, G., Yin, J., Guan, H., Lv, L., ... & Li, J. (2020). Research activity does not affect nest predation rates of the Silver-throated Tit, a passerine bird building domed nests. *Avian Research*, 11(1), 28.
7. 胡骞, 臧少平, 徐克阳, 薛琳, 王希明, 丁长青, & 李建强. (2019). 山东发现褐翅燕鸥和白腰燕鸥. *动物学杂志*, 4.
8. Wang, H., **Hu, Q.**, Wang, Y., Lv, L., Wang, P., Zhang, Z., & Li, J. (2026). Density and kinship interactively affect natal dispersal in a social bird. *Proceedings of the Royal Society B: Biological Sciences*, 293(2070), 20252985.
9. 李寒晴, 刘纯焱, 黄晓芳, 李淳滢, 胡骞, 王悦, 李建强, 闫利平, & 张东. (2026). 金光喜鸟虱蝇线粒体基因组及系统发育分析(双翅目:虱蝇科). *昆虫分类学报*, 48(1), 58-70.
10. 李新宇, 刘正霄, 胡骞, 彭于杨, 王辉, 李建强, ... & 徐基良. (2025). 吐哈盆地鸟类多样性的初步研究. *干旱区地理*, 48(10), 1707-1720.
11. Peng, P., Chen, W., Qianhan, N., Yu, H., **Hu, Q.**, Zhu, J., ... & Li, J. (2025). Hanging Position of Artificial Nest Boxes Affects Reproductive Success of Russet Sparrow *Passer cinnamomeus*. *Animals*, 15(10), 1413.
12. Jin, T., **Hu, Q.**, Liu, Y., Li, J., Pan, S., & Xu, J. (2025). Latitudinal variations in incubation behaviour among uniparental incubating birds. *Journal of Animal Ecology*, 94(4), 800-810.
13. Wang, Y., **Hu, Q.**, Wang, Y., Liu, J., Du, Z., Xu, J., & Li, J. (2024). Selective effect of winter weather on a songbird's morphology depends on individual sex and winter condition. *Oecologia*, 205(2), 339-350.
14. Jin, T., Lu, S., Wang, Y., Hua, J., Liu, Z., **Hu, Q.**, ... & Xu, J. (2024). The clutch size, incubation behavior of Reeves's Pheasant (*Syrnaticus reevesii*) and their responses to ambient temperature and precipitation. *Avian Research*, 15, 100168.
15. Wang, Y., **Hu, Q.**, Xu, J., & Li, J. (2023). Sex-specific selective effect of winter weather on morphological traits in a small passerine bird. *Avian Research*, 14, 100093.
16. 朱家贵, 黄涛, 杜志勇, 胡骞, & 李建强. (2023). 营巢植物影响红头长尾山雀的巢被捕食风险. *动物学杂志*, 58(3), 330-340.
17. Lu, S., Liu, Z., Tian, S., Song, K., **Hu, Q.**, Li, J., & Xu, J. (2022). Sex-specific movement responses of Reeves's pheasant to human disturbance: Importance of body characteristics and reproductive behavior. *Animals*, 12(13), 1619.
18. 张颖, 黄婷婷, 胡骞, & 朱建宁. (2022). 基于鸟类多样性提升的社区公园生境 营造策略探析. *中国园林*, 38(3), 106.
19. 尹江南, 余高阳, 努尔索力坦·恰汗, 胡骞, 王悦, & 李建强. (2021). 大山雀身体大小的性二态及其季节变化. *动物学杂志*, 56(6), 10.
20. 石江艳, 华俊钦, 胡骞, 蒋淼, 赵玉泽, & 徐基良. (2020). 河南董寨国家级自然保护区发现灰头鹀. *动物学杂志*, 55(3), 305-305.

HONORS & SCHOLARSHIPS

2024	One-year Visiting PhD Scholarship from the China Scholarship Council (CSC)
2022	Outstanding Student Cadre
2021	Excellent master thesis (school level)
2017	Excellent bachelor thesis (school level)
2026	National Encouragement Scholarship, China
2015	Outstanding Student Cadre

PRESENTATIONS & POSTERS

2026	Presentation	Parental care and bird embryonic metabolism: a comparison between uniparentally incubating Silver-throated Tits and biparentally incubating Black-throated Tits	20 th International Society for Behavioral Ecology Congress	Turin, Italy
2023	Presentation	两种长尾山雀的不同孵卵模式及其适应性的比较研究	第八届中国林业学术大会	哈尔滨
2022	Presentation	环境温度变化对两种具有不同孵卵策略鸟类孵卵投入的影响	第 16 届翠鸟论坛	线上
2021	Poster	两种长尾山雀卵壳图案变化规律的研究	2021 年北京动物学会学术研讨会	北京
2019	Poster	研究者访巢活动对两种长尾山雀巢存活率的影响的研究	第 15 届中国鸟类学大会	长春
2019	Presentation	研究活动会影响鸟类的巢存活率吗？——以两种长尾山雀为例探究查巢频率对鸟类巢存活率的影响	2019 年北京动物学会学术研讨会	北京

REFERENCES

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